

mCLM: A Function-Infused and Synthesis-Friendly Modular Chemical Language Model

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Abstract

Despite their ability to understand chemical knowledge and accurately generate sequential representations, large language models (LLMs) remain limited in their capacity to propose novel molecules with drug-like properties. In addition, the molecules that LLMs propose can often be challenging to make in the lab. To more effectively enable the discovery of functional small molecules, LLMs need to learn a molecular language. However, LLMs are currently limited by encoding molecules from atoms. In this paper, we argue that just like tokenizing texts into (sub-)word tokens instead of characters, molecules should be decomposed and reassembled at the level of functional building blocks, i.e., parts of molecules that bring unique functions and serve as effective building blocks for real-world automated laboratory synthesis. This motivates us to propose mCLM, a modular Chemical-Language Model tokenizing molecules into building blocks and learning a bilingual language model of both natural language descriptions of functions and molecule building blocks. By reasoning on such functional building blocks, mCLM guarantees to generate efficiently synthesizable molecules thanks to recent progress in block-based chemistry, while also improving the functions of molecules in a principled manner. In experiments on 430 FDA-approved drugs, we find mCLM capable of significantly improving 5 out of 6 chemical functions critical to determining drug potentials. More importantly, mCLM can reason on multiple functions and improve the FDA-rejected drugs (“fallen angels”) over multiple iterations to greatly improve their shortcomings. †

1 Introduction

Small molecules—the class of chemical matter primarily built from carbon atoms bonded together—can perform a wide range of important functions in human society. These include essentials like promoting health by acting as medicines [125, 20, 81, 90, 110], converting energy by functioning as key components in solar cells [69, 65, 48, 49, 79], and achieving sustainability by serving as inherently recyclable products. These functions also include many nice-to-haves that drive substantial economic growth, including acting as colorants, flavorings, perfumes, cosmetics, lotions, sunscreens, coatings, quantum dots, molecular computers, chemical biology probes, insect repellants, etc.

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†All codes, data, and models will be released publicly upon publication.

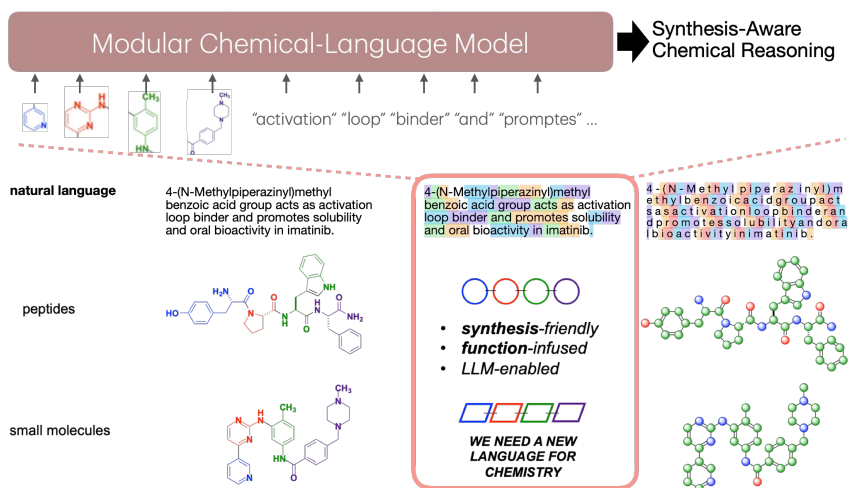


Figure 1: mCLM adopts a modular chemical vocabulary, which uses molecular *modules* as tokens together with natural language tokens. Note that we also refer to these modules as “building blocks”. Compared with using natural language names, SMILES strings, or holistic embeddings for whole molecules, this level of tokenization aligns better with molecular function groups. It allows more efficient molecular structure modification and synthesis.

A major challenge with small molecules, however, is that the traditional approach for small molecule synthesis is highly artisanal, slow, and expensive. It considers all possible arrangements of atoms and centers on creating for each targeted molecule a unique process for its assembly that leverages a menu of thousands of different reactions each run under thousands of possible conditions and using millions of possible starting materials. This approach imposes several limitations: (1) It is unfriendly to automation – machines are not good at doing thousands of different things thousands of different ways from a million different starting points. (2) It leads to an undemocratized landscape in drug discovery. With development costs averaging around \$1.3 billion per drug, only economically advanced countries can afford to invest in such high-risk research [41]. Moreover, participation in the process of molecular innovation currently requires access to highly trained experts in chemical synthesis. (3) Most importantly, there are many instances of known commercial drugs or materials that have well-documented limitations that have remained unaddressed. For example, Imatinib is an important anticancer drug that works well in most parts of the body, but it only poorly penetrates the blood-brain barrier [87, 77, 14, 36]. This means it may effectively treat cancer at its primary site, yet fail to prevent fatal brain metastasis. In such cases, more blood-brain-barrier penetrant molecules that retain all of the other favorable properties of the current drug are desired, but making such selective modifications in functional properties can be very challenging. As another example in the materials domain, organic photovoltaic (OPV) molecules are expected to be more economical and environment-friendly alternatives to current solar cells. However, current commercialized OPV devices either achieve less than 10% energy conversion efficiency—significantly lower than traditional silicon solar cells—or have stability far short of 10 years [82]. Overcoming these types of drawbacks of current molecules remains a challenging problem in the domain.

An alternative block-based chemistry approach for small molecule synthesis has recently emerged [28, 107, 50, 45, 92, 8, 4, 100, 84], and we recognized that this approach provides an opportunity to reimagine the intersection of AI and small molecule science. Block-based chemistry is chemistry that machines can do. It iteratively assembles small molecules from prefabricated building blocks using chemistry that is simple and general and thus readily automated. Akin to automated DNA, RNA, and peptide synthesis platforms, a major strength of this block-based approach is that it can access billions of novel small molecules with high degrees of functional potential using only a few automation-friendly reactions and a bounded set of pre-fabricated function-infused building blocks.

We specifically recognized that this block-based approach provided an opportunity to create a new modular language for chemistry. The idea that molecules and their synthesis may be best understood from the perspective of a “chemical language” dates back to 2014 [11], where structural fragments and functional groups play the roles of “chemical words”. This view is supported by multiple observations aligned with natural language: (1) they can be decomposed and reassembled, (2) they

exhibit ambiguity—the same building block can perform vastly different functions depending on the chemical context, and (3) they possess significant diversity, as many different structures can lead to the same function. This linguistic parallel suggests the potential to train a large language model specifically for molecules by linearizing their structures into modular sequences.

A common representation for such an approach is the Simplified Molecular Input Line Entry System (SMILES) [104], where atoms are denoted by one- or two-character symbols (e.g., C for carbon, Br for bromine, and F for fluorine), rings are represented with numbers, and branches are indicated using parentheses. However, such atom-level tokenization strategies (e.g., SMILES [104] or SELFIES [43]) resemble character-level natural language models, which struggle to generalize effectively. Unlike proteins, which have a fixed vocabulary of 20 amino acids, small molecules exhibit an open vocabulary when each atom is treated as a token, as illustrated in Figure 1. It also causes severe restrictions in practicality because many of the new structures proposed by LLMs based on atom-level tokenization are not practically synthesizable in the laboratory. This approach has created a major gap between what is now possible in silico and what is possible in the physical world.

Furthermore, the SMILES representation can obscure critical structure information: two atoms that are direct neighbors in the molecular graph may be distantly separated in the SMILES string. Even with recent efforts to align SMILES and natural language description [17, 73, 2], integrate chemical properties and functional groups [71] into SMILES, incorporate 3D geometric information [25, 54], and employ graph neural networks to capture molecular graphs and chemical reaction contexts [97], these representations still fail to encapsulate functional knowledge that is often described only in natural language literature because the inherent properties and functions of molecules are hidden in their structure, composition, and interaction.

Against this backdrop, we argue that there is a correctable fundamental mismatch between the way LLMs work and the way chemists traditionally synthesize and study small molecules. Increasing the “tokenization granularity” down to the corresponding letters is not helpful, and in fact makes the extraction of meaning much more complex and causes generative AI to hallucinate words that don’t exist. To bridge this gap, in this paper, we aim to teach computers to speak two complementary languages: one that represents molecular building blocks (i.e., subgraph structures) indicative of specific functions and compatible with automated modular assembly, and another that describes these functions in natural language (Figure 2). Unlike existing approaches that add such knowledge as a post hoc step, we developed a function- and synthesis-aware modular chemical language model (mCLM). Inspired by bilingual speakers who frequently “code-switch” (naturally and often switch between their two languages within the same message), we propose a novel neural encoder that integrates molecular structure and natural language. mCLM incorporates both function- and synthesis-related knowledge into the small molecule tokenization process a priori.

First, we tokenize small molecules at the level of building blocks (graph substructures) that are better able to predict function and are, by design, fully compatible with automated modular small molecule synthesis. We use graph neural networks to encode each building block. We then extract natural language sentences from the literature that describe the molecule’s functions and chemical reactions and synthesis constraints of various molecules, and seamlessly insert the encoded building blocks alongside the corresponding entity names, as illustrated in Figure 2. During the inference stage, mCLM enables these functional modules to be predictably and automatically assembled into new molecules with desired functions.

mCLM has multiple potential advantages of encoding and generating molecules at a modular level:

1. **Synthesis efficiency:** synthesis can be faster and more broadly accessible because the process is, by design, simple, iterative, general, and machine-friendly. This can enable rapid iterative drug and material development and even automated lab experiments.
2. **Alignment with language models:** resembling the mechanism of natural language tokenization, molecular modularization provides a more natural interface to align with word representations.
3. **Reasoning:** leveraging LLMs’ instruction following capability and wide-scale pretraining, mCLM is able to iteratively refine molecules based on knowledge of molecular functions.

2 The Modular Chemical Language Model

2.1 Overview

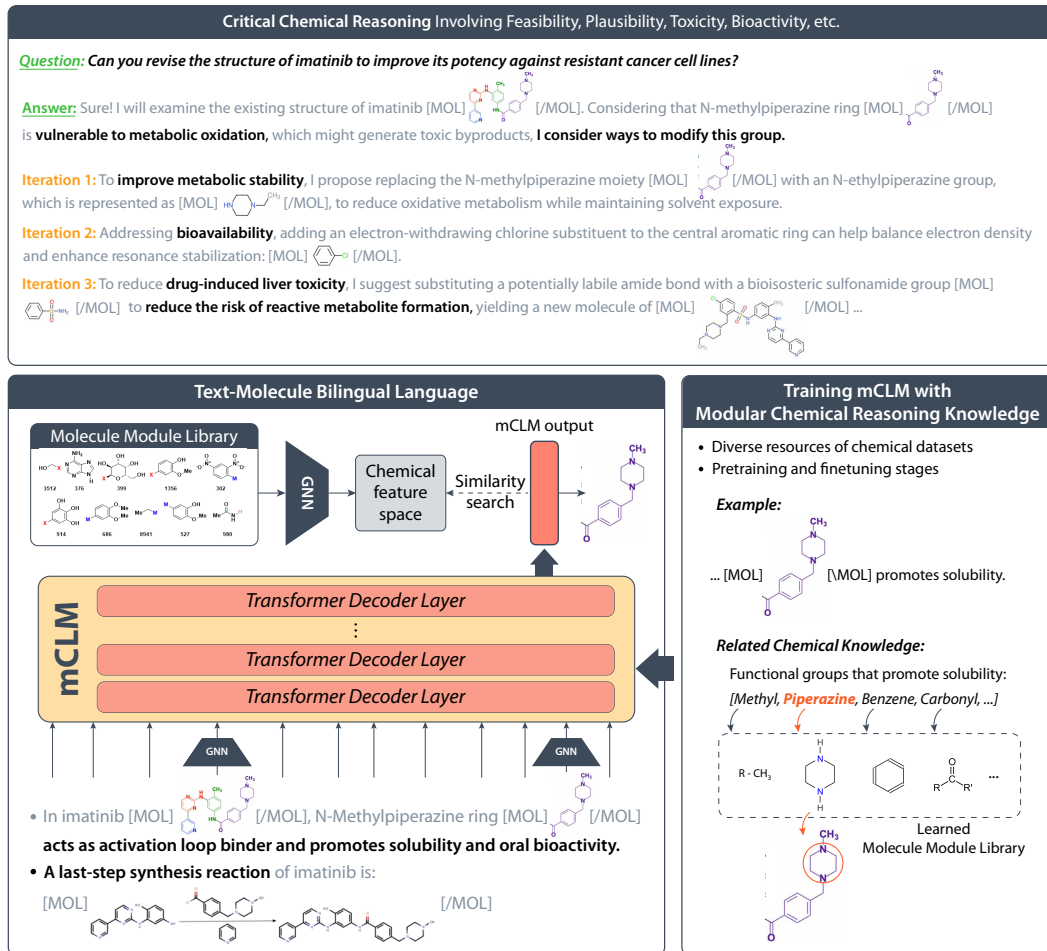


Figure 2: An overview of the mCLM. It is a multimodal chemical-language model that tokenizes molecules into *building blocks*. After being trained on datasets consisting of properties, functions and synthesis data, the mCLM can conduct critical chemical reasoning through an iterative process to refine molecular design.

Small molecules pose unique challenges to tokenize and encode due to their perceived open-ended structural diversity. This has caused the field to consider them differently from natural language where well-defined vocabularies exist. In natural language modeling, tokenization identifies common substrings (such as words and subwords), which carry richer semantic information than sequences of characters. We note that most small molecules that occur in nature are similarly composed primarily of connected *building blocks* [45, 92]. There is likewise a high degree of inherent modularity in many medicines and materials [23, 22, 3, 5, 45, 92, 95]. We propose mCLM, a multimodal model that jointly encodes and understands natural language and molecules based on synthesis-friendly building blocks instead of atoms. In Section 2.2, we introduce the concept of molecular building blocks as a chemical “vocabulary” in addition to the natural language vocabulary we use. Section 2.3 then describes the tokenization process to obtain the library of building blocks. Then in Section 2.4 we describe how the mCLM trains and runs on the token sequences. Finally, we introduce the reasoning mechanism of mCLM which refines molecule design over multiple iterations in Section 2.5.

2.2 A Function-Infused and Synthesis-Friendly Vocabulary

In this work, we propose to leverage a chemical vocabulary of synthesis-friendly building blocks. This approach guarantees capacity for automated iterative assembly [51]. The blocks can be chemically connected using predefined synthesis rules in a short period. In particular, we propose a chemically meaningful vocabulary, V . Briefly, akin to language models developed for peptides/proteins, small molecules can be assembled automatically from accessible building blocks. These building blocks are often highly associated with chemical functions, such as binding to protein targets, modulating enzyme activity, or affecting involved metabolic processes. This will enable rapid and iterative proposal of new small molecules, automated synthesis of those small molecules, and generation of the corresponding functional data on demand. Notably, in contrast to SMILES strings which break molecule structure during graph linearization (e.g., separating physically adjacent subgraphs such as the two carbons in molecule $C(N)C$), our representation is linear in the physical world.

2.3 Tokenization

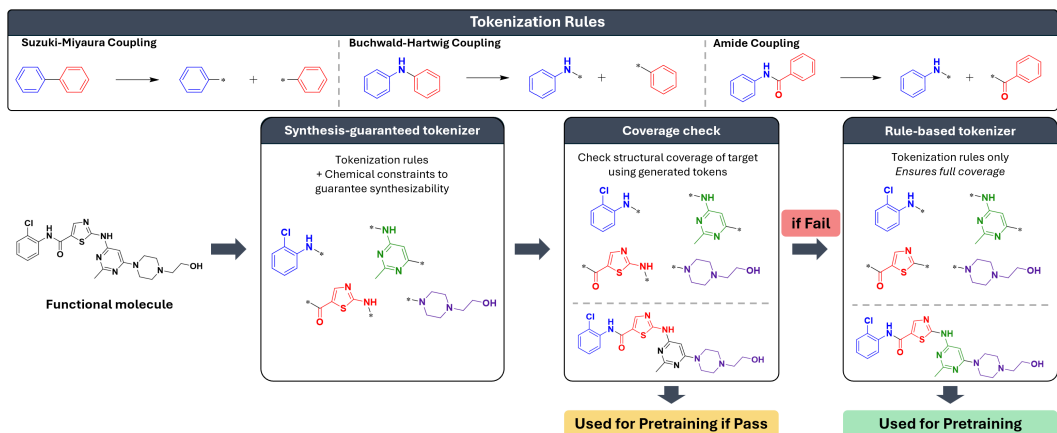


Figure 3: An overview of the tokenization process. A functional molecule is first processed by the synthesis-guaranteed tokenizer to produce a set of building blocks compatible with automated modular synthesis. These blocks are then evaluated via a structure coverage check to determine whether they fully reconstruct the original molecule. If coverage is complete, the blocks are used directly for pretraining. Otherwise, the molecule is reprocessed using a rule-based tokenizer to ensure full representation for training purposes.

To generate molecules that are not only valid but also automatically synthesizable, we tokenize molecular inputs into molecular “building blocks” (i.e., subgraphs) that align with known chemical synthesis procedures. During training, we combine two tokenization strategies to balance coverage and synthetic feasibility: a synthesis-guaranteed tokenizer and a rule-based tokenizer (Figure 3). The synthesis-guaranteed tokenizer disconnects the molecule only at bonds that can be formed by a predetermined small set of reactions that can be performed in an automated manner. Specifically, it only permits bond disconnections that correspond to reactions compatible with state-of-the-art automated synthesis platforms. These three bonds are: amide coupling, Suzuki-Miyaura coupling, and Buchwald-Hartwig coupling (see Figure 3 tokenization rules) [93]. Unlike traditional retrosynthesis tools that prioritize maximum coverage or human-designed heuristics, this tokenizer is guided by the operational constraints of block-based chemistry. It also ensures that resulting blocks are free from functional group conflicts that would interfere with downstream synthesis. Please see Appendix D.1 for more information. This conservative approach leads to a set of molecular tokens that are guaranteed to work under defined synthesis protocols, enabling seamless transition from model-generated output to real-world synthesis. When the synthesis-guaranteed tokenizer cannot fully cover a molecule—due to chemical incompatibilities or reaction constraints—we fall back on a rule-based tokenizer to ensure coverage of a larger variety of molecules. The rule-based tokenizer also breaks molecules along the same automated machine-friendly bonds as the above tokenizer. However, no other rules are applied beyond specifying a minimum size for blocks. This rule-based tokenizer

is used only during training to support learning on diverse molecular structures while maintaining consistency with synthesis logic.

2.4 Chemical-Language Modeling

Figure 2 illustrates the architecture of mCLM. The model is a sequential generative model that processes molecule and text sequences in a unified manner. It adopts a Transformer architecture, which is well-suited for handling sequential data and allows for using pre-trained language models as a backbone. After tokenizing the molecules as mentioned in the last section, we encode each building block using graph neural networks (GNNs) [19, 83, 26]. These representations are then concatenated with natural language embeddings at positions where molecule entity names appear. This results in a form of “code-switched” language which blends molecular structures with natural language descriptions. The feature sequence is then fed into a Transformer decoder-only architecture, which predicts the next token based on previous tokens. This allows for pre-training the model on a large corpus of multi-modal data, enabling it to learn and “talk about” the relationships between the different modalities and their respective representations.

We train this model—mCLM—on top of open-source large language models pretrained on general-domain corpora, allowing us to leverage their linguistic capabilities without incurring the computational cost of training from scratch. As the training objective, we adopt a unified categorical cross-entropy (CCE) loss applied to both natural language and molecular tokens:

$$\mathcal{L} = H(P(\mathbf{X}), P_{\theta}(\mathbf{X})), \quad \text{logit}(P_{\theta}(v \mid \mathbf{X}_{1 \dots i-1})) = \begin{cases} \mathbf{c}^{\top} \mathbf{e}_v, & v \in \mathcal{V}_{\text{natural language}} \\ \mathbf{c}^{\top} f_{\theta}(\text{GNN}_{\theta}(v)), & v \in \mathcal{V}_{\text{molecular building block}} \end{cases}$$

Specifically, the loss is computed between the ground truth distribution $P(\mathbf{X})$ and the model-predicted distribution $P_{\theta}(\mathbf{X})$ over the combined vocabulary of natural language and molecular building blocks. The model generates logits for the next token v by computing the dot product between the contextual representation $\mathbf{c}$ with token embeddings. $\mathbf{c}$ is produced by the Transformer given the previous tokens $\mathbf{X}_{1 \dots i-1} = [X_1, \dots, X_{i-1}]$. For natural language tokens, the embedding $\mathbf{e}_v$ is directly taken from the pretrained natural language model. The embedding for molecular building blocks is computed by passing the building block’s graph through a GNN, followed by a linear adapter function f_{θ} to project it into the same embedding space. This formulation enables joint training over both modalities using a single loss function. More detailed training procedure is described in Appendix C.

2.5 Critical Chemical Reasoning

Chemical reasoning over molecules often involves optimization of multiple functions, such as toxicity, bioactivity, binding affinity, etc. Therefore, it is not a straightforward task to propose an ideal molecule structure, especially with only a single attempt. In nature, organisms evolved over billions of years to produce complex molecules that are optimal for their use. Additionally, many functions positively or negatively correlate with each other. Optimizing one function may lead to trade-offs in others. For example, many drugs with higher potency were rejected due to increased toxicity to patients. To address this, we propose a reasoning process that allows the model to refine its own generated molecules and iteratively improve their desired functions. At each step, mCLM proposes a modification of the molecule for improvement by considering one or more functions and self-evaluates the effect of the proposed modification. Then in the next iteration, mCLM identifies those functional building blocks that still have room for improvement. This process is repeated until a maximum number of iterations is reached, or if mCLM fails to find additional reasonable modifications. As illustrated in Figure 2, this process can be summarized in the following algorithm:

3 Experimental Evaluation

3.1 Creating Oracle Models for Evaluation

To evaluate the performance of our generated molecules, we construct oracle models focused on Absorption, Distribution, Metabolism, Excretion, Toxicity (ADMET) property prediction. We select 6 tasks from the Therapeutics Data Commons (TDC) benchmark [33]: **AMES** (mutagenicity), **BBBP** (blood-brain barrier permeability), **CYP3A4** inhibition (metabolism), **DILI** (drug-induced liver injury), **HIA** (human intestinal absorption), and **PGP** (P-glycoprotein substrate classification).

Algorithm 1 Critical Chemical Reasoning for Molecule Design in mCLM

```
1: Input: Initial molecule  $M_0$ 
2: Output: Refined molecule  $M^*$ 
3: Initialize  $M \leftarrow M_0$ 
4: while True do
5:   Enumerate over functions to improve in  $M$  (e.g., metabolism, drug-induced organ injury,
   blood-brain barrier penetration)
6:   if No clear objective remains for improvement then
7:     return  $M$ 
8:   end if
9:   mCLM generates a candidate modification  $M'$  by replacing, adding, or removing building
   blocks in  $M$ 
10:  mCLM evaluates  $M'$  with respect to desired functions
11: end while
```

Models are trained using predefined scaffold-based data splits with an 8:1:1 ratio for train, validation, and test sets. The detailed training procedure is described in Appendix B.2.3. We opt to use robust foundation models—FARM [71], ChemBERTA-2 [2], and a GNN [19]—for ensemble learning. To build the ensemble, we use each model as a feature extractor. The extracted features are concatenated and passed through a fully connected layer for final prediction.

3.2 Improving FDA-Approved Drugs with Out-of-Vocabulary Blocks

As a large-scale test, we apply the mCLM to improve all FDA-approved drugs consisting of at least 3 blocks. This amounts to 430 molecular structures and 796 unseen blocks. Since most of these molecules (426/430) contain blocks that were not present in the 1,000 blocks used for training, this presents an opportunity to find how the mCLM performs out-of-distribution. Results in Table 1 show that improvement is achieved for 5/6 properties, even though the model has not seen almost half of the blocks in its vocabulary during training.

Property	FDA Drug	Modified
AMES Mut. (↓)	59.5	54.0
CYP3A4 Inhib. (↓)	2.0	1.2
BBBP (↑)	37.6	41.4
HIA (↑)	93.2	97.6
DILI (↓)	66.2	55.5
PGP (↓)	66.0	68.0

Table 1: Average pharmacokinetic and toxicity properties of FDA drugs with 3 or more blocks and their proposed modifications. (↓: lower is better, ↑: higher is better; $p < 0.05$ by Wilcoxon Test [105, 74].)

3.3 Case Study: Multi-step Critical Reasoning to Resurrect the “Fallen Angels”

In a similar vein, there are many new drug candidates that almost make it to FDA approval but fall short for various reasons when being tested in clinical trials. For example, Evobrutinib is a Bruton’s tyrosine kinase (BTK) inhibitor that went through clinical trial as a drug for relapsing Multiple Sclerosis. However, the FDA placed a partial clinical hold on Phase III trials in April 2023 after two patients showed signs of drug-induced liver injury [67]. TNG348 is a USP1 inhibitor designed for treating BRCA1/2-mutant and HRD cancers, but it failed in phase 1/2 clinical trials due to liver abnormalities [35, 80]. These “fallen angels” represent tremendous opportunities for impactful engagement of the AI/chemistry interface, because much is known about the strengths of each of these small molecules, and it is also known why they fell short. Fixing such fallen angels is a high leverage opportunity for the function-infused and synthesis-friendly mCLM to contribute. Herein we demonstrate this potential with some real-world examples of fallen angels that can be optimized with mCLM. Importantly, and uniquely, all of the new compounds proposed by mCLM are, by design, well-suited for rapid automated assembly using robots.

Figure 4 shows an application of the mCLM to these two molecules. For both, the initial step is to optimize DILI, the reason the drugs failed in clinical trials. Following that, the mCLM is applied to fix other properties which were made worse in the previous attempt (PGP for Evobrutinib and BBBP for TNG248). For good measure, another property of each molecule is then improved. Notably, at each

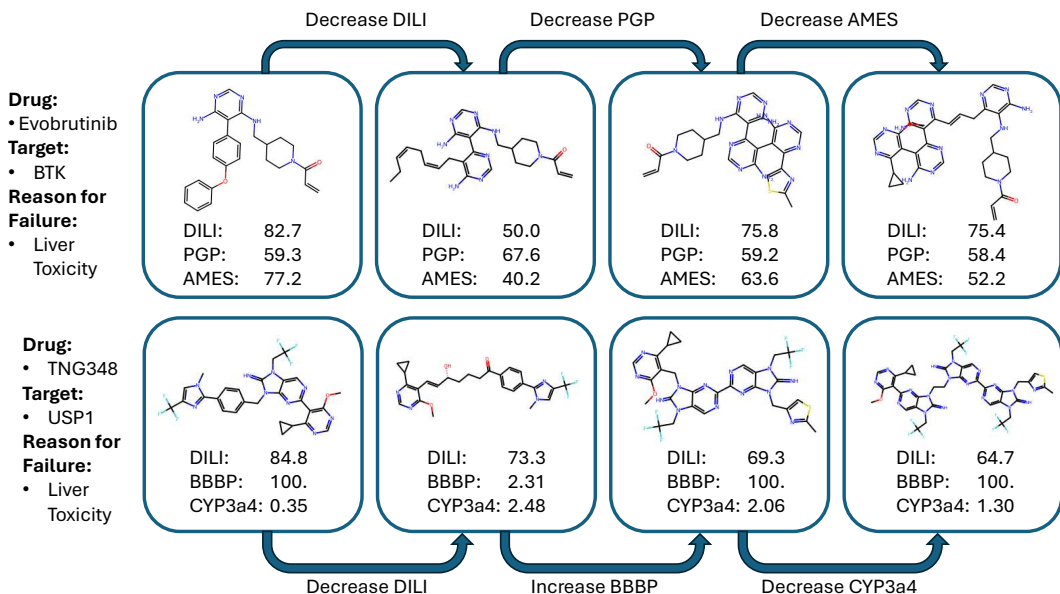


Figure 4: Examples of fallen angel property modification.

step, the mCLM only makes minor modifications to each drug of roughly 1 building block. While the mCLM shows promising results for repairing these drugs, it is worth noting that drug discovery is a many-objective optimization problem. While we are able to generate proposed molecules with improved toxicity relative to Evobrutinib and TNG348, as well as other properties, yet other important properties may still have been compromised. Future work may want to investigate longer reasoning chains across a wider variety of properties.

3.4 Library Design: Identifying Functionally-Important Blocks

One notable thing about the architecture of the mCLM is the ability to identify blocks which are preferred for certain functions. This can help inform virtual screening campaign design (e.g., which 10 blocks should we use to get the best chemical space for BBBP?), and it can also be useful for stimulating scientific inquiry. As an example, Figure 5 shows the most frequent modifications preferred by the mCLM for improving DILI in FDA drugs. Interestingly, these modifications tend to be small changes from the original molecule block.

4 Related Work

4.1 Applying LLMs to Chemical Representation and Discovery

Following their success in general natural language processing tasks [123], large language models (LLMs) have been increasingly adopted for solving a wide range of problems in computational chemistry [116, 119]. Existing efforts can be broadly categorized into two groups based on molecular representation: (1) SMILES-based modeling [24, 61, 72, 124], and (2) graph-based modeling [59]. For SMILES-based approaches, MolT5[18] introduces a multimodal language model

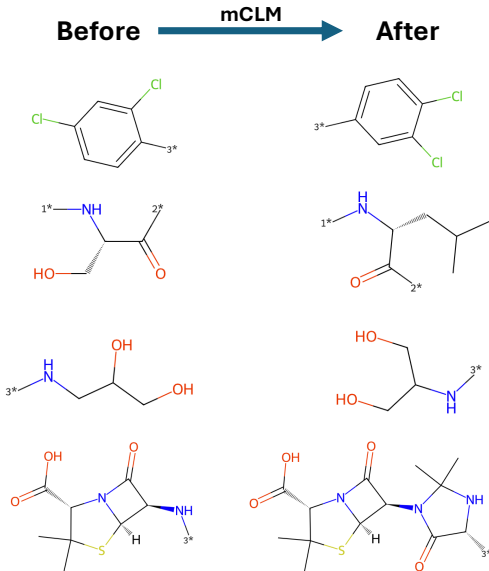


Figure 5: Four of the most frequent mCLM-proposed modifications to improve DILI in FDA-approved drugs.

that jointly understands and generates textual and SMILES representations. GIMLET[122] unifies graph and textual information using a distance-aware graph-text Transformer. LlaSMol[115] is instruction-tuned for property prediction and molecular description. DrugAssist [114] explores interactive molecule optimization by enabling LLMs to perform goal-driven modifications through human-in-the-loop dialogue. Graph-based approaches aim to capture molecular structure more explicitly. MolCA [59] integrates graph encoders with Galactica [89] using a cross-modal projector. 3D-MoLM[53] extends this paradigm to 3D molecular structures by incorporating a 3D molecular encoder. Llamole[55] further advances this direction by enabling interleaved generation of text and molecular graphs, supporting applications such as retrosynthetic planning and molecular inverse design. However, most molecule language models tokenize at the level of atoms [99, 2, 126, 109] and/or they do not infuse synthesis considerations up front, which precludes matching tokens with desired functions and/or leads to AI-generated new molecular targets that are not readily synthesizable.

4.2 Functional Groups/Fragments

Functional groups/fragments are core structural motifs that govern molecular reactivity, biological activity, and physicochemical properties, forming the basis of chemical understanding [23, 75, 39]. Their study has propelled advances in medicinal chemistry, natural product research, and material science by guiding synthesis and performance optimization [23, 22, 3, 5].

Given their interpretability and functional relevance, functional groups/fragments are also increasingly being incorporated into machine learning models for molecular representation [46, 70, 29, 38, 68, 102, 118, 120]. Some approaches employ a masking strategy for masked atom prediction by simultaneously masking all atoms within a functional group, applied to both molecular graph and SMILES representations [103, 46]. Other approaches include augmenting SMILES strings with functional group annotations [68], building functional group-level graphs [68], or leveraging hierarchical architectures that model both atomic-level and functional module-level representations [29, 70].

Our work goes beyond mere chemical representation by attempting to encode molecules at the building-block level within natural language LLMs, paving the way for chemical reasoning. These methods also often overlook the synthesizability of molecules. They tend to break molecules at bonds that are difficult to form through chemical reactions, resulting in generated molecules that are challenging—or even impossible—to synthesize using standard reaction pathways. This gap limits the practical applicability of the learned representations, especially for tasks such as molecular generation or optimization.

4.3 Multimodality, Multilinguality and Code-Switching in LMs

This work is also inspired by exploration in enabling LLMs to understand multiple languages [34, 10, 117, 15, 91] or data modalities such as vision and audio [52, 1, 57, 64, 98, 96, 113, 13, 101, 12, 88, 111]. [32] fed genome sequences to LLMs and found them capable of suggesting common functions for gene sets while also providing interpretable rationales. Explorations on proteins [60, 78] show success in protein-to-text and text-to-protein generation in applications including antibody design and protein understanding. This work is also related to other recent studies that have explored the integration of graph neural networks (GNNs) as feature vectors into LMs capture both textual and structural information [59, 122, 37].

5 Conclusions and Future Work

In this work, we propose the modular Chemical-Language Model, which is the first attempt to jointly model natural language sequences with modular chemical language. mCLM is capable of conducting critical chemical reasoning by iteratively refining molecule designs over diverse objectives and chemical knowledge. Future work could incorporate richer information such as 3D structures of molecules and physical constraints into mCLM representation. Other modalities such as protein and nucleic acid sequences will further enable mCLM to reason on biological activity, protein docking knowledge, and individuals’ genetic profiles. Future efforts could also extend chemical reasoning to additional aspects such as collecting and leveraging results from physical simulation tools, including protein interaction dynamics, retrieval from chemical and reaction knowledge bases, etc. Another important direction is to introduce multi-agent interactions with other AI agents, human scientists, and automatic laboratories, where proposed molecules can be efficiently synthesized using modular chemistry technologies. There is also great potential in including alternative scientific hypothesis

generation and plausibility prediction. In the long-term, we envision a landscape where mCLMs can be included in a more comprehensive chemical research workflow consisting of fully automated and never-ending reasoning - proposal - synthesis - physical testing - feedback - reasoning loops. mCLM can serve as a preliminary step in bridging AI and physical world experiments, shattering the barriers that currently preclude non-specialists in organic synthesis from meaningfully participating in small molecule innovation.

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A Pretraining Data Mixture

To train the model, our goal is to cover a wide variety of different molecular functions and classes of molecules. To do so, we consider a wide variety of data sources. We use three existing instruction datasets: SMolInstruct [115], Tulu-3 [44], and the Biomolecular text portion of Mol-Instructions [24]. SMolInstruct is used to cover the standard set of chemistry LLM tasks. We supplement this with the non-overlapping portion of Mol-Instructions. Finally, we include Tulu-3 to preserve general reasoning capability.

A.1 Property-Based Synthetic Instruction Data Generation for Contrastive Learning

Additionally, we augment this data mixture with a significant amount of synthetic data created using real-world property datasets. Given our datasets, we create additional question datapoints for both regression and classification tasks, such as “Is this molecule <SMILES> ... </SMILES> blood brain barrier permeable? Yes”. We also consider the opposite direction: property to molecule and multiple properties to molecule, and unconditional molecule generation. The templates used for constructing this data were created using GPT-4o; 50 were originally generated and then bad templates were removed by hand. Further, we augmented each data point with a description of the property, also written by GPT-4o.

In addition to these tasks, we also consider the molecule optimization task as described in DrugAssist [114] (e.g., given molecule A, improve property X). However, their approach has fundamental limitations due to the reliance on oracle models. These models may be out-of-distribution for molecules within our dataset, and low-quality models may propagate errors into our training data. To address this issue, we consider “activity cliffs” [85, 121] within existing property datasets to train the model for molecule optimization. Activity cliffs are two molecules which are structurally similar but have large differences in a given property. We opt to base our definition of activity cliff on Murcko scaffolds [7]. This methodology, which is widely used in medicinal chemistry, extracts the core subgraph, or scaffold, of a molecule. We found that this approach was much more computationally feasible on large-scale molecule datasets than approaches like fingerprint similarity [6].

Given nominal data (e.g., toxic, kinase inhibitor, banana smell), we look for a pair of molecules that have the same scaffold but only one molecule has the desired property. Given numerical data (e.g., solubility, power

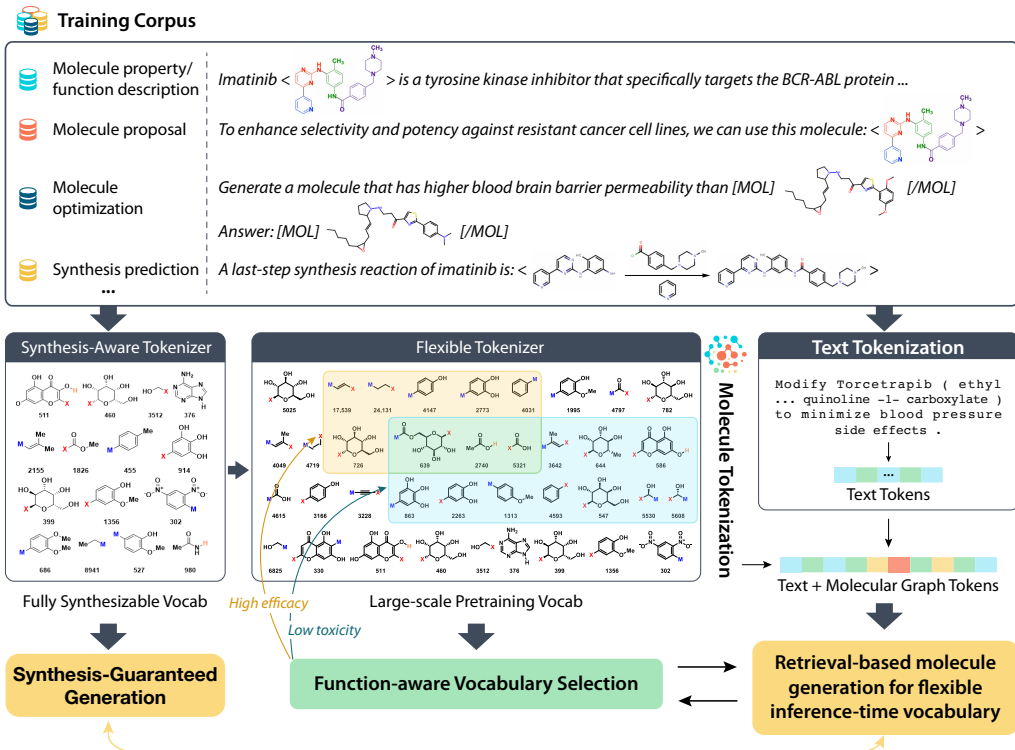


Figure 6: Overview of Data Creation.

conversion efficiency, HOMO-LUMO gap), we instead look for molecules which have a property difference of 1 standard deviation. This forms the positive-negative same-scaffold portion of our data. We also consider pairs of molecules that have different scaffolds but the same property. Here, our goal is to teach the model to consider function over structure. As an example, we may ask “Propose a molecule with a different structure than [MOL] ... [MOL] that still demonstrates anticoagulant properties. [MOL] ... [MOL]”. For a select number of property types, we use oracle models instead of ground truth data. Please see Appendix B.2.3 for more information.

A.2 Data Filtering

Due to the large size of our dataset, we filter our data to only keep the most frequent 100k (50k end blocks and 50k mid blocks), 10k (5k and 5k), and 1000 blocks (500 and 500) for experiments. This is because there are far fewer molecule tokens compared to text (each training sample only has up to about 10). Due to Zipf’s law [127], in the full dataset, a single molecule token may only appear in one example in the entire training corpus. Without considerably scaling the compute budget, understanding this additional block is difficult. Overall, this filtering enables us to learn more efficiently and requires less memory resources for training the model, and allows us to better test the model architecture. Please see the section on training details for more information in Appendix C and see Appendix E for source property datasets.

Data Type	Full Data	Only Blocks	Top 100k	Top 10k	Top 1000
All	36,641,170	20,910,431	14,823,280	6,195,903	1,054,124
Existing Data Sources	3,117,841	1,208,365	1,049,473	537,674	149,994
SMolInstruct	2,124,738	215,262	56,370	14,242	2,300
Tulu-3	939,343	939,343	939,343	469,672	93,934
Mol-Instructions Biomol. Text	53,760	53,760	53,760	53,760	53,760
Synthetic Real Data	21,290,353	9,294,434	7,711,548	3,960,633	705,210
Classification	5,612,215	3,771,038	3,192,583	1,337,619	223,341
Molecule Generation	535,317	316,630	245,034	150,001	27,393
Positive Negative Same Scaffold	3,795,820	113,390	65,566	16,286	2,007
Positive Positive Different Scaffold	660,925	477,976	256,435	191,720	33,579
Property to Molecule	6,676,957	3,125,097	2,878,012	1,639,322	307,780
Multi-Property to Molecule	572,505	556,107	533,647	423,089	78,793
Scaffold+Property to Molecule	705,833	405,497	226,696	80,448	13,078
Regression	2,730,781	528,699	313,575	122,148	19,239
Synthetic Oracle Data	12,232,976	10,407,632	6,062,259	1,697,596	198,920
Classification	2,077,327	1,778,874	1,058,754	299,444	35,168
Positive Negative Same Scaffold	72,763	58,645	25,035	4,935	469
Scaffold+Property to Molecule	1,169,407	925,055	453,087	113,049	13,642
Property to Molecule	13,257	13,254	8,841	906	2
Regression	8,900,222	7,631,804	4,516,542	1,279,262	149,639

Table 2: Dataset categories and their respective sample counts.

Subset	Source	Molecules				Molecule Tokens			
		Total	Tokenized	Untokenized	Scaffolds	Cap	Mid	Total	Synthesis-Aware
Full Data	SMolInstruct	1,951,205	1,566,030	385,175	510,464				
	Our data	6,160,565	5,270,982	864,212	1,109,562				
	Total	7,994,305	6,787,879	1,178,957	1,537,424				
Only Blocks	SMolInstruct	459,910	459,910	0	145,614	157,204	59,681	216,885	50,788
	Ours	3,830,543	3,830,543	0	705,641	531,472	175,903	707,375	189,770
	Total	4,220,604	4,220,604	0	799,267	598,470	203,810	802,280	214,431
Top 100k	SMolInstruct	146,079	146,079	0	49,743	22,871	14,809	37,680	10,930
	Ours	2,345,519	2,345,519	0	429,960	49,941	49,780	99,721	27,823
	Total	2,458,034	2,458,034	0	457,032	50,000	50,000	100,000	28,220
Top 10k	SMolInstruct	37,686	37,686	0	12,373	4,130	2,534	6,664	2,618
	Ours	821,238	821,238	0	147,524	5,000	4,998	9,998	3,468
	Total	848,674	848,674	0	153,144	5,000	5,000	10,000	3,487
Top 1000	SMolInstruct	4,867	4,867	0	1391	477	359	836	420
	Ours	109,554	109,554	0	19,639	500	500	1,000	432
	Total	112,657	112,657	0	20,101	500	500	1,000	432

Table 3: Breakdown of molecule data.

B Synthetic Data Generation

B.1 Task Formulation for Molecular Design

To train our modular chemical language model (mCLM), we generate synthetic data that reflect realistic scenarios encountered in molecular design. These include tasks relevant to drug discovery and organic photovoltaic (OPV) materials design. Our goal is to equip the model with the ability to understand and reason over molecular representations and natural language prompts across a diverse set of tasks.

Drug Discovery Tasks: In the context of drug discovery, chemists often engage in iterative and multi-objective optimization processes, where molecules are evaluated, modified, or generated based on various physicochemical and pharmacological properties. These tasks typically involve querying for specific properties, modifying structures to meet certain design criteria, or generating novel candidates that satisfy given constraints. We consider the following tasks that a chemist may perform:

- **Text-to-Molecule Generation:** Generate a molecule from a textual description of its structure or properties.
- **Property Prediction:** Given a molecule and a property of interest, predict whether the molecule possesses the property (binary classification) or the quantitative value of the property (regression).
- **Molecular Optimization:** Modify a given molecule to satisfy or improve a specific property.
- **Scaffold-Constrained Generation:** Given a scaffold and a target property, generate a molecule that satisfies both the structural constraint and the property constraint.

OPV Material Design Tasks: To support broader applications of mCLM beyond drug discovery, we also incorporate tasks relevant to organic photovoltaic (OPV) material design. OPVs are an emerging class of lightweight, flexible materials used for solar energy harvesting, where the power conversion efficiency (PCE) is the primary performance metric. A typical OPV device is composed of a donor molecule and an acceptor molecule. The donor is responsible for absorbing sunlight and generating excitons (electron-hole pairs), while the acceptor facilitates charge separation and electron transport. The chemical compatibility and electronic alignment between the donor and acceptor molecules critically influence the resulting PCE. To enable learning in this domain, we define the following OPV-specific tasks

- **PCE Prediction:** Predict the PCE of a given donor-acceptor pair.
- **Donor/Acceptor Completion:** Given a donor (or acceptor) and a target PCE, generate the complementary component (acceptor or donor) that achieves the desired performance.
- **Constrained Completion with Scaffold:** Generate donor or acceptor molecules that match a given scaffold and achieve a target PCE.

B.2 Instruction Tuning Data Generation

B.2.1 Prompt-Answer Templates

To generate instructional data, we construct a pool of question and answer templates for each task. These templates include multiple paraphrased variants to introduce linguistic diversity and improve the model’s generalization capability. During data generation, a question template is randomly sampled from the question set and populated with sample-specific content. Likewise, a corresponding answer template is sampled from the answer set to form a complete prompt-response pair. For example:

- **Question Template:** “Given [a molecule] and [a property of interest], modify the molecule to achieve [desired property value].”
- **Answer Template:** “The [property] of [molecule] is [value].”

This templating strategy allows us to produce a large number of diverse, semantically equivalent training instances that support instruction tuning across multiple molecular design tasks.

B.2.2 Label Sources for Instruction Tuning Data

To enable diverse and meaningful pretraining for mCLM, we incorporate both experimentally derived and model-generated labels, covering a broad spectrum of molecular properties critical to chemistry, pharmacology, and materials science. This dual-labeling strategy allows the model to learn from abundant low-level molecular descriptors while also reasoning over high-level functional and biological endpoints.

Low-Level Molecular Properties from ChEMBL: For foundational chemical descriptors, we leverage the ChEMBL25 database—a comprehensive bioactivity resource containing approximately 2 million compounds

with rich structural and physicochemical annotations. ChEMBL25 serves as an abundant and reliable source of labels for low-level properties that are widely used in cheminformatics pipelines. From this corpus, we select a core set of descriptors that are most informative for molecular design: Hydrogen bond acceptors (HBA), Hydrogen bond donors (HBD), LogP (octanol–water partition coefficient), Molecular weight (MolWt), Number of aromatic rings, Number of rotatable bonds, Topological polar surface area (TPSA). These descriptors are inexpensive to compute and provide critical insights into molecular solubility, permeability, and synthetic feasibility—making them essential for early-stage screening and property-based filtering.

High-Level Molecular Properties via Oracle Labeling: In addition to low-level properties, we aim to expose the model to high-level functional endpoints that capture complex biological phenomena. Such endpoints are central to pharmacokinetics, drug safety, and efficacy, but they are rarely available in large quantities due to the high cost of experimental validation. Consequently, labeled datasets for these tasks are limited in size and diversity. To address this challenge, we employ the oracle ensemble models to generate synthetic labels for a curated set of ADMET tasks.

B.2.3 Ensemble Oracle Model:

To generate high-quality synthetic labels for downstream tasks, we construct oracle models focused on ADMET property prediction. We select tasks from the Therapeutics Data Commons (TDC) benchmark [‡] using the following criteria:

- **Relevance to Drug Discovery:** The task must reflect a critical aspect of drug efficacy or toxicity.
- **Predictability:** The task must be reliably predictable using existing models. Specifically, we evaluate all 22 ADMET-related classification tasks in TDC and retain only those where standard models achieve an area under the ROC curve (AUC) greater than 0.80. This ensures the synthetic labels are sufficiently accurate for training purposes.

Based on these criteria, we select six tasks:

- **AMES** (mutagenicity),
- **BBBP** (blood-brain barrier permeability),
- **CYP3A4** inhibition (metabolism),
- **DILI** (drug-induced liver injury),
- **HIA** (human intestinal absorption),
- **PGP** (P-glycoprotein substrate classification).

TDC provides predefined scaffold-based data splits with an 8:1:1 ratio for train, validation, and test sets. This splitting strategy ensures that structurally dissimilar compounds are separated across subsets, encouraging generalization to novel scaffolds.

Although TDC provides leaderboards for these tasks, many top-performing entries lack reproducible code or working implementations. For instance, the authors of one of the top submissions explicitly acknowledge on GitHub that their code is not runnable.[§] Therefore, we opt to use robust foundation models—FARM, ChemBERTA-2, and a GNN—for ensemble learning.

- **FARM** [68]: A SMILES-based BERT model trained with functional group-aware tokenization.
- **ChemBERTA-2** [2]: A large-scale transformer model trained on millions of canonical SMILES sequences.
- **GNN** [19]: A graph neural network trained on molecular graphs with atom- and bond-level features.

To build the ensemble, we use each model as a feature extractor. The extracted features are concatenated and passed through a fully connected layer for final prediction. This ensemble approach is stacking, where multiple base learners feed into a meta-learner. For each task, we select a threshold that maximizes the F1 score on the validation set. This threshold is then used to binarize the predicted logits into class labels. The performance of our ensemble model across the selected tasks is summarized in Table 4.

[‡]https://tdcommons.ai/benchmark/admet_group/overview/

[§]<https://github.com/maplightrx/MapLight-TDC>

Table 4: Performance (AUC) of individual models and the ensemble across six selected ADMET tasks.

Model	AMES	Pgp	DILI	BBBP	CYP3A4	HIA
FARM [68]	0.88	0.89	0.79	0.94	0.88	0.92
GNN [19]	0.75	0.78	0.86	0.79	0.80	0.81
ChemBERTa-2 [2]	0.86	0.89	0.81	0.93	0.86	0.99
Ensemble	0.89	0.91	0.84	0.93	0.89	0.99

C Training Procedure

We employed Qwen2.5-3B [112] as the starting LLM for building the mCLM. Generally, we followed the training procedure from LLaVa [56, 58, 47]. We used a two-layer MLP with PReLU activation [30] as an adaptor into the LLM input/output from the GNN. We selected an initial learning rate of 1e-5 for the full model and 1e-6 for the adaptor and LM heads. Further, we used a cosine annealing schedule with a minimum of 1e-6 and 2000 linear warmup steps; AdamW [63] optimizer was employed. The model was trained on 4 A100 80GB GPUs in bfloat16 precision.

We found that the model learned the molecule tokens much slower than the text (there is usually a 10x difference in loss value). Molecule tokens are rarer and show up less in the training data. Because of this, we decided to separate the molecule classifier head and the language classifier head. We used a standard autoregressive language modeling loss for both, and we averaged these two losses for the final loss value. The main part of our training experiments focused on minimizing the molecule loss, since the text loss was easy to optimize. Further, we found PEFT [31] was not sufficient to adapt to molecules, so full finetuning was required. Roughly 10-50 examples from each synthetic (data source, task) pair were put into a validation set.

To initialize the GNN weights, we employed the MolCLR [103] unsupervised contrastive learning technique. We used AugliChem [66] for the augmentations: random atom masking, random bond deletion, and motif removal. One of these augmentation was selected uniformly at random for each data point. The GNN was initialized using a batch size of 128 and lr of 1e-4 with a cosine schedule. The model was trained on all 800k blocks in the full data until convergence on a validation set. Please see Appendix ?? for vocabulary examples and their nearest neighbor embeddings. We tested embedding dimensions between 16 and 4,096 and found 128 dimensions to be sufficient while minimizing total memory cost. This was necessary because we stored the entire embedding matrix in GPU memory, which was much faster, but consumed about 20GB VRAM. Doing so allowed us to train without the GNN during our pretraining process, which is considerably more efficient. We note the GNN can then be finetuned along with the rest of the mCLM during finetuning to new types of molecules or specific tasks. While we did consider a sampled softmax to train the mCLM, we found this to limit the learning of the model.

For training the mCLM, we used two stages for pretraining. First, we trained for 1 epoch with everything frozen except the adaptor, to allow the adaptor to adjust to the LLM representation space. For the second stage, we trained for 5 epochs with only the GNN embeddings frozen. As discussed in the training data mixture section A, we used the most frequent 1000 building blocks as our vocabulary.

After pretraining, we finetune the mCLM to standardize its outputs for our experiments. During pretraining, we train for robustness by using a wide variety of responses (e.g., for BBBP prediction we might respond "It is restricted from entering the central nervous system" instead of 'No'). For finetuning, we train with standardized responses for our desired tasks (e.g., "Generate a molecule that has higher blood brain barrier permeability than [MOL] ... [MOL].", "[MOL] ... [MOL]"). Due to our downstream tasks, we finetuned exclusively on the molecule optimization task for 5 epochs over 100k examples for each property. We trained using the same procedure as the pretraining stage, but we selected the best model using validation loss.

D Tokenizer Details

D.1 Synthetic Tokenizer Details

The synthesis-guaranteed tokenizer disconnects the molecule only at bonds that can be formed by a pre-determined small set of reactions, preferably only those that can be performed in an automated manner. For instance, if amide bond formation is defined as available, the tokenizer will be able to disconnect amide bonds in the molecule of interest. Up to this point, the protocol is synonymous with classical computational retrosynthesis but there is a fundamental difference. Namely, the sets of reactions suitable for automated synthesis is very limited – in fact, state-of-the-art synthesis machines utilize only three types of disconnections (amide bond formation, as well as Suzuki and Buchwald-Hartwig couplings). This places very stringent requirements on the groups that can be present in the disconnected blocks – for instance, when the disconnection (say, Buchwald-Hartwig coupling) yields an amine functionality on one of the blocks, this block cannot contain any groups that during the anticipated uses of this block would present a synthetic incompatibility. In the most trivial case, the block cannot contain another unprotected amine because after the Buchwald-Hartwig disconnection, the block would feature two amines which, in turn, would present competing reactive sites (in the Buchwald-Hartwig synthesis but also in the formation of amide bonds). Therefore, the tokenizer performs retrosynthetic operations while simultaneously checking if they do not lead to blocks with functional groups presenting competing reactivities. Only disconnections avoiding such problems are allowed. This then guarantees that when the corresponding blocks are used to make other molecules, they give only the selective synthesis outcomes.

E Source Datasets and Databases

- Leffingwell odors [76]
- BACE [108]
- Flashpoint [86]
- MUV [108]
- Tox21 [108]
- AMES [33]
- Bioavailability [33]
- Caco2 [33]
- Carcinogens [33]
- cav3_t [33]
- choline_transporter [33]
- clearance hepatocyte [33]
- CYP1A2 [33]
- CYP2C9 [33]
- CYP2D6 [33]
- CYP3A4 [33]
- DILI (drug induced liver toxicity) [33]
- Half_life [33]
- hERG [33]
- HIA (human intestinal absorption) [33]
- Hydration free energy [33]
- kcnq2 [33]
- LD50 [33]
- Lipophilicity [108]
- m1_muscarinic [33]
- OPV data [69, 62]
- orexin_receptor [33]
- PAMPA_NCATS [33]
- Broccatelli [9]
- potassium_ion_channel [33]
- PPBR (Plasma Protein Binding Rate) [33]
- Pubchem logP [40]
- SARSCoV2_3CL [33]
- SARSCoV2_vitro [33]
- serine_threonine_kinase_33 [33]
- Skin Reaction [33]
- Solubility_AqSolDB [33]
- tyrosyl-dna_phosphodiesterase [33]
- VDss (volume of distribution at steady state) [33]
- Molecule Property Cliff Datasets (30+ datasets) [94]
- Chemical Function (CheF) [42]
- ChemFOnt: the chemical functional ontology resource [106]
- Pubchem properties [40]
- FreeSolv [108]
- QM8 [108]

- QM9 [108]
- Thermosol [108]
- ESOL (Estimated SOLubility) [108]
- Lipo [108]
- BBBP [27]
- ClinTox [108]
- HIV [108]
- SIDER [108]
- Forward synthesis (USPTO) [115]
- Retrosynthesis [115]
- CheBI-20 [16]
- L+M-24 [21]
- HBA [27]
- HBD [27]
- MolWt [27]
- NumAromaticRings [27]
- rotatable_bonds [27]
- TPSA (topological polar surface area) [27]